MACHINE LEARNING • API • DEPLOYMENT

Adult Census Income Prediction

Completed end-to-end project: model experiments, MLflow tracking, API, interactive interface, and Docker deployment.



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1. Business problem and objective

A repeatable workflow turns census-style input data into an accessible income-category prediction.

THE PROBLEM

Large datasets can be difficult to interpret manually. Analysts need a consistent way to examine patterns associated with income categories.

THE SOLUTION

A trained machine-learning model predicts whether annual income is ≤$50K or >$50K from the input features.

THE VALUE

Predictions are accessible through a web interface and API, while experiments remain inspectable in MLflow.

Scope and responsible use

- Predictions are estimates based on learned patterns—not proof of an individual's actual income.
- Demographic data can carry bias; assess fairness and avoid using predictions as the sole basis for high-impact decisions.

2. Model development and evaluation

Candidate experiments and recorded metrics are available in the local MLflow dashboard.

01

Prepare data

Load, inspect and preprocess the Adult Census data.

02

Train models

Run experiments and compare candidate configurations.

03

Evaluate

Review recorded metrics and plots to understand performance.

04

Package

Package the model and preprocessing artifact for inference.

METRICS TO REVIEW

Accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix. Report values from actual runs.

EVIDENCE AVAILABLE

MLflow experiments and runs are visible and available to inspect recorded metrics, parameters and artifacts.

3. Test-set performance results

Actual metrics recorded in the MLflow run supplied for this presentation.

Accuracy

83.25%

Overall proportion of test examples classified correctly.

Precision

62.16%

Of predicted >$50K cases, about 62% were correct.

Recall

77.93%

The model identified about 78% of actual positive-class cases.

F1-score

69.16%

Balances precision and recall.

ROC-AUC

90.47%

Strong ability to rank positive cases above negative cases.

Average precision

77.41%

Summarises the precision–recall trade-off across thresholds.

WHAT THE RESULTS SUGGEST

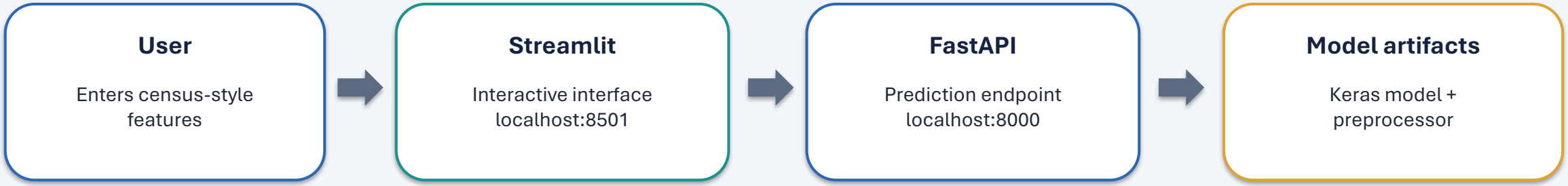
Accuracy is 83.25% and ROC-AUC is 90.47%, indicating useful overall discrimination on the test set.

BUSINESS TRADE-OFF

Recall (77.93%) is higher than precision (62.16%): the model catches many positive cases, but some predictions will be false alarms.

3. Deployed application architecture

The API and interactive frontend run together through Docker Compose.



VERIFIED LOCALLY

Docker Compose starts both containers; the API health endpoint reports the model and preprocessor loaded.

ACCESS POINTS

Streamlit: localhost:8501 • API docs: localhost:8000/docs • Health: localhost:8000/health

MLflow tracks experiments separately; the running API loads packaged artifacts from the application folder.

4. Completion status and handover

The core workflow is implemented and verified locally.

Model experiments	Experiments, runs and metrics are visible in MLflow.
Prediction API	FastAPI health check and prediction endpoint tested.
User interface	Streamlit frontend runs and calls the API.
Containerisation	Docker Compose starts API and frontend together.
Version control	Project committed and pushed to a new branch.

THANK YOU